

# Optimal design of a hybrid photovoltaic–wind power system with the national grid using HOMER: A case study in Kerkennah, Tunisia

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**Abstract**—Renewable energy is certain to play a key role in future electricity generation due to the rapid depletion of conventional energy. Photovoltaic and wind energy are the major renewable energy sources. However, renewable energies are an inexhaustible, expensive, and unpredictable source of energy. An alternative solution is to combine one or more renewable energy with other conventional energy. In recent years, the research interest towards the utilization of hybrid energy systems in desalination plants. This paper aims to optimize several hybrid energy system models consisting of photovoltaic, wind, and the national grid in desalination plant in Tunisia. Optimization is based on the techno-economic analysis of the proposed energy system is performed by using HOMER simulation software. The simulation will be focused on the net present costs, Levelized cost of energy, produced excess electricity, the Renewable Fraction of Energy, and the reduction of CO<sub>2</sub>emission for the hybrid energy configurations. Results show that the system photovoltaic, wind, and the national grid is the best energy system installed in the desalination plant.

**Keywords**—HOMER, solar PV, wind, national grid, desalination.

## I. INTRODUCTION

### A. Background and Research Motivation

In 2020, according to statistics from the International Energy Agency, more than three quarters of the world's energy consumed was of fossil origin. However, these sources of energy (gas, oil, coal) condemned to dry up over the next few decades, have proven today to be the main cause of air pollution and the increase in greenhouse effect causing global warming [1]. Global energy demand rose sharply rapidly after the Great Depression of the 1930 and continues to grow. Fig.1 represents the total World Energy Consumption.

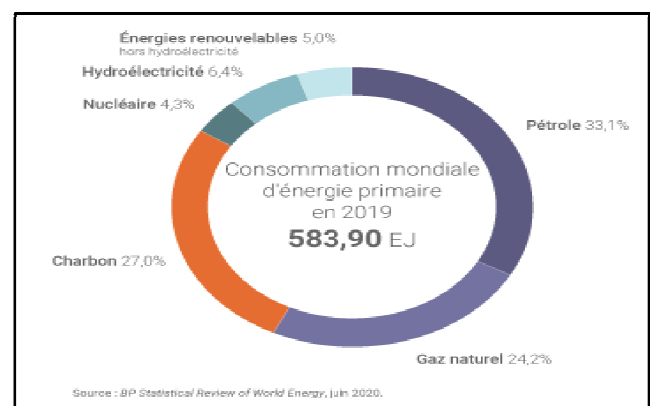


Fig.1 Total World Energy Consumption [1]

Saltwater desalination has been a process evolving for 30 years. The number of these stations now exceeds 17,000 stations with a capacity of 100 million m<sup>3</sup> per day. Each installation can produce a few liters to several hundred thousand m<sup>3</sup> of water per day [2]. The desalination process requires a large amount of energy to produce freshwater and the overwhelming majority of the energy currently used for desalination is obtained from fossil fuels [1]. The use of renewable energies for desalination remains limited [3]. Photovoltaic and wind energy are the most promising alternative to conventional energy because it is more available and sustainable than other renewable energy [4].

### B. Literature Overview

Desalination is a large consumer of energy, thereby there are various studies on simulation on the optimization of energy systems in the desalination project. Nowadays, a number of simulation software has been developed for the optimization of energy systems, among which the best known are: HOMER, HYBRID2, EnergyPLAN, SOMES, TRNSYS16, HOGA, and RETScreen. Table 1 shows a brief summary of the studies published using software tools for optimization energy system.

Table 1

State of the art of photovoltaic-wind hybrid system with the national grid by simulation

| Ref. | software  | Energy Sources |     |    |    |    | Best energy system |
|------|-----------|----------------|-----|----|----|----|--------------------|
|      |           | WT             | SPV | FF | BT | GC |                    |
| [5]  | HOMER     | X              |     | X  |    |    | WT+FF              |
| [6]  | HOMER     | X              | X   |    |    | X  | WT+SPV+GC          |
| [7]  | HOGA      | X              | X   | X  | X  |    | WT+SPV+FF+BT       |
| [8]  | TRNSYS    | X              | X   |    | X  |    | SPV+BT             |
| [9]  | HOMER     | X              | X   |    | X  |    | SPV+BT             |
| [10] | HOMER     |                | X   |    |    | X  | SPV+GC             |
| [11] | HOMER     |                | X   | X  | X  |    | SPV+FF+BT          |
| [12] | HOMER     | X              | X   | X  | X  |    | WT+SPV+FF+BT       |
| [13] | RETScreen | X              | X   |    | X  |    | WT+SPV+BT          |
| [14] | HOMER     | X              | X   |    | X  |    | WT+SPV+BT          |
| [15] | HOMER     |                | X   |    |    | X  | SPV+GC             |
| [16] | HOMER     | X              | X   | X  | X  |    | WT+FF+BT           |
| [17] | HOMER     | X              | X   | X  | X  |    | SPV+FF+BT          |
| [18] | HOMER     | X              | X   |    | X  | X  | WT+SPV+GC          |
| [19] | HOMER     | X              | X   | X  | X  |    | WT+SPV+BT          |
| [20] | HOMER     |                | X   | X  | X  |    | SPV+FF+BT          |
| [21] | HOMER     | X              | X   | X  | X  | X  | WT+SPV+FF          |
| [22] | HOMER     | X              | X   | X  | X  |    | WT+SPV+FF+BT       |
| [23] | HOMER     |                | X   | X  | X  | X  | SPV+GC             |
| [24] | HOMER     |                | X   |    |    | X  | SPV+GC             |
| [25] | HOMER     | X              | X   |    |    | X  | WT+SPV+BI+GC       |
| [26] | HOMER     |                | X   |    | X  | X  | SPV+GC             |
| [27] | HOGA      | X              | X   |    | X  |    | WT+SPV+BT          |
| [28] | HOMER     |                | X   | X  | X  |    | SPV+BT             |
| [29] | HOMER     | X              | X   | X  | X  |    | SPV+BT             |
| [30] | HOMER     | X              | X   |    | X  |    | WT+SPV+BT          |

WT: Wind Turbine; SPV: Solar Photovoltaic; FF: Fossil Fuel; BT: Battery; GC: Grid-connected.

It can be seen that the HOMER software has been widely used in studies described in the table. HOMER software developed in the United States by National Renewable Energy Laboratory. This software makes it possible to carry out simulations to optimize hybrid energy systems, not connected to the national grid, for isolated sites [5--9] ; [11--14] ; [16] ; [17] ; [19] ; [20] ; [22] ; [27--30] or connected to the national grid [6] ; [10] ; [15] ; [18] ; [21] [23--26].

### C. Objective of the Study

The recent literature on the subject includes a number of studies that implement the simulation for a sustainable energy source selection. However, the objective of this paper is to optimize several hybrid energy system models consisting of solar PV, wind, and the national grid in a desalination plant in Tunisia. This paper used HOMER simulation software to reduce the cost of electrical energy production and reduction in greenhouse gases.

## II. MATERIALS AND METHODS

### A. Study Area

Tunisia is a fossil fuel-producing country mainly by gas power plants and its climate is conducive to the production of renewable energy sources, thus in Tunisia, energy studies are very limited.

A statistical analysis of the distribution of Tunisian energy consumption shows that pumping stations and water production stations, established by the National Water Exploitation and Distribution Company (SONEDE), are the largest consumers of energy based on fossil fuels with 463 GWh [31--36].

### B. Proposed approach

Generally, the installation of renewable energies requires optimal sizing. Climatic changes in the site concerned influence the sizing of the hybrid energy system. We have applied HOMER software to size a set of energy systems. It is based on the net present cost (NPC) of the project, which is the overall cost of installation and the cost of operation of the system including the initial cost of installation, the cost of replacement, the cost of operation, and maintenance during its lifetime. This paper aims to optimize renewable energy connected to the national grid in a desalination plant in Kerkennah (Tunisia).

### C. Application and discussions

The HOMER software makes it possible to introduce information concerning the various components (photovoltaic, wind turbine, batteries, and grid connection), the costs of the components, the availability of resources based on energy demand, and operational life. Thus, it is possible to analyze several different configurations for this same energy system in order to obtain an economically optimized model. Fig.2 represent the input data are the geographical position, electrical demand, energy sources and the parameters economic technical, environmental of HOMER software.

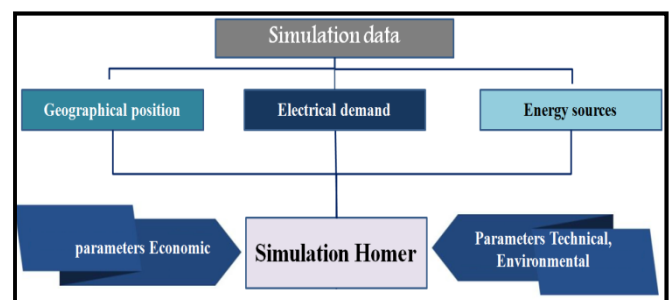


Fig.2 The input data of HOMER software

Faced with the growing demand for water in the world and more particularly in Tunisia, unconventional water resources, and especially water desalination, have been the subject of strong development for two decades. Tunisia's experience in desalination is quite rich insofar as it began with the desalination of brackish groundwater in 1983 with the implementation of a 3300 m<sup>3</sup>/day station in Kerkennah with the reverse osmosis membrane technique. The desalination station of Kerkennah is located between 34°42'N (north position) latitude and 11°13'E (east-west position) longitude. The cost of electricity represents 35.5% of the cost per cubic meter of water produced. It constitutes the largest part of the cost of desalination. Reducing this cost is a major issue in the station. Fig.3 represents the different costs of the Kerkennah desalination plant.

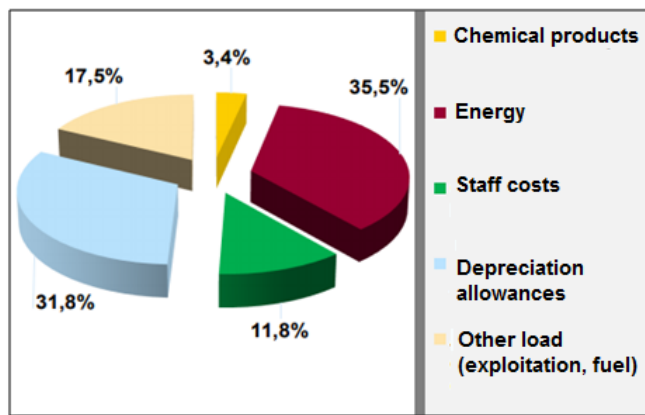


Fig.3 Cost distribution in the Kerkennah station [37]

The load consumption is considered as 5707kWh/d. The power consumption of the desalination plant is stable from day to day and from season to season. This is due to the fact that work at the industrial level is regular (24 hours a day). Weekends and holidays have no influence on power consumption, as shown in Fig. 4 and Fig. 5.

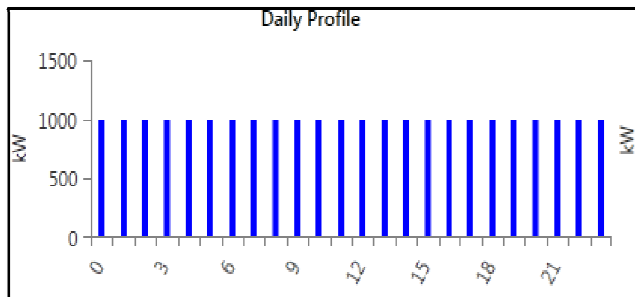


Fig. 4 Day-to-day load profile of the Kerkennah desalination plant

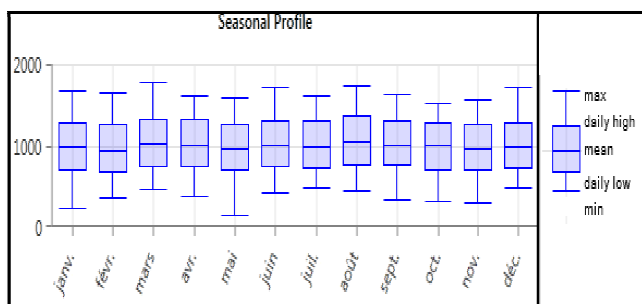


Fig. 5 Monthly load profile of the Kerkennah desalination plant

The data of wind speed and solar radiation is obtained from NASA surface meteorology and solar energy database. After simulation, the results obtained by HOMER presents the technical and economic details such as the net present cost, the cost of energy in £/kWh, the percent of renewable fraction, the percent of excess energy, and greenhouse gas emissions in kg/year. Table 2 represents the main results obtained by the simulation from the HOMER.

Table 2  
Results obtained after simulation

| Alternatives | NPC (£)     |                       | COE (£) | fraction of renewable energy(%) | excess of renewable energy (%) | Carbon dioxide (kg/y) | Nitrogen oxide (Kg/y) |
|--------------|-------------|-----------------------|---------|---------------------------------|--------------------------------|-----------------------|-----------------------|
|              | Capital (£) | Exploitation cost (£) |         |                                 |                                |                       |                       |
| SPV+WT+BT+GC | 2484546     | 13093129              | 0,149   | 53,7                            | 19,6                           | 417,996               | 48,3                  |
| SPV+WT+GC    | 2501469     | 13078181              | 0,149   | 53,8                            | 19,9                           | 416,979               | 48,1                  |
| SPV+GC       | 2179776     | 16707555              | 0,181   | 38,2                            | 22,9                           | 557,473               | 64,4                  |
| SPV+BT+GC    | 2283672     | 16629952              | 0,181   | 38,6                            | 24,6                           | 553,740               | 63,9                  |
| WT+GC        | 1000000     | 18553723              | 0,187   | 31,5                            | 0,6                            | 618,118               | 71,4                  |
| WT+BT+GC     | 1012921     | 18583468              | 0,188   | 31,5                            | 0,6                            | 618,118               | 71,4                  |
| GC           | 0           | 26201659              | 0,251   | 0                               | 0                              | 901,963               | 104,0                 |

The simulation by the HOMER software shows that the system includes photovoltaic panels with a total power of 559 kW, two wind turbines with a unit power of 100 kW, 5 batteries of 2.51 kWh and converters of 282 kW and the national grid is the best energy system in desalination plant in Kerkennah fig. 6. Currently, the unit cost of electricity is £ 0,251/kWh by the Tunisian National Grid (STEG). The simulation results showed that the system PV/Wind/Battery/National Grid production cost is £ 0,149 /kWh which is quite low compared with conventional energy (National Grid).

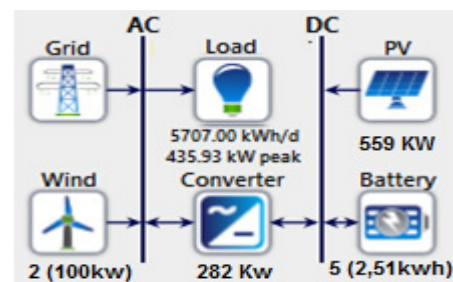


Fig. 6 PV/Wind/Battery/National Grid systems

### III. CONCLUSION

The energy situation in Tunisia is marked by limited resources based on fossil fuels, a decrease in production, and a sharp increase in demand.

The use of renewable energies is a major concern for major energy consumers. These have economic, technical, social and environmental advantages compared to conventional energies such as oil, gas, coal, etc. Today, we find on the market a wide variety of renewable energy sources such as solar and wind. In this paper, we have used HOMER software to optimize several energy system models in a desalination plant in kerkennah. After simulation the hybrid configuration PV/Wind/Battery/National Grid is the best economic system with a net present cost.

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